Deceptive Opinion Detection Using Stacking-Based Deep Ensemble Learning

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Abstract—The widespread reliance on online reviews in e-commerce and social platforms has led to an increase in deceptive opinions, commonly called fake reviews or opinion spam. These reviews manipulate consumer perceptions, impacting purchasing decisions, brand credibility, and platform trustworthiness. Detecting fake reviews is challenging due to their human-like writing style, evolving deceptive tactics, and the scarcity of well-annotated datasets. Traditional machine learning models, such as Support Vector Machines (SVM) and Naïve Bayes, struggle to achieve high accuracy. In contrast, deep learning models like CNN and LSTM perform better but still face limitations in capturing diverse deceptive patterns. This study proposes a stacking-based deep ensemble learning model that integrates Simple Flatten, CNN, LSTM, and BiLSTM as base classifiers with a logistic regression meta-learner for improved classification performance. Evaluated on the Amazon fake review dataset, the proposed model achieves an accuracy of 90.04%, outperforming individual deep learning models such as BiLSTM (87.13.00%), LSTM (86.10%), CNN (83.11%) and MLP (80.00%). This will provide a more reliable and scalable solution for online platforms to detect deceptive opinions.

Index Terms—Fake News, Machine Learning, Transformer, DistilBERT, Transformer, Support Vector Machine

I. Introduction

With the rapid growth of e-commerce and social platforms, users increasingly rely on online reviews for decision-making. Platforms like Amazon and Yelp help consumers assess products and services based on previous customers' opinions [1]. Since buyers cannot examine products in person, they rely on these reviews to assess the overall quality and credibility. However, the growing importance of reviews has increased fake reviews or opinion spam, which consists of fabricated opinions that are expressed to deceive consumers [2]. Businesses often create fake positive reviews to improve their image, while competitors undermine other businesses through fake negative reviews to tarnish their images. These misleading reviews, to say the least, alter the consumers' decisions, incur losses, and damage faith in online systems. It is hard to tell whether reviews are legitimate or fake because it is difficult to differentiate the writing styles. Unlike the type of spam that exists in bulk, fake reviews are meticulously designed to blend in. There are insufficient large, deeply annotated datasets because many existing ones are built using synthetic or learnt heuristics, making detection much harder. With the aid of bots and organised humans, deception has escalated, and so has the technology to bolster it [3]. The credibility of fake reviews is deceitfully increased by using fake likes, upvotes, and synchronised review campaigns.

Furthermore, detecting fake reviews gets even more challenging as they are often strengthened by fake engagement, such as likes and upvotes. For effortless and efficient detection of these agendas, a multi-method approach that integrates the text's content, the reviewers' conduct and the posting actions must be employed [4]. Support Vector Machines (SVMs) [5] and Naïve Bayes classifiers fake review detection is done with traditional machine learning models, but their performance is often lacking in accuracy. Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) [6] networks do better but still have their issues. For example, CNNs may not be good at retrieving long-range dependencies in the text.

On the other hand, LSTMs, though great for sequential data, do not capture the whole range of deceptive opinions and the context behind them. These weaknesses show the gaps that are still left unfilled. Fortunately, approaches like stacking-based deep ensemble learning help by combining the strengths of multiple models. It is a semi-algorithmic process. Multilayer Perceptron (MLP) with CNN, LSTM, and BiLSTM integration improve feature extraction, model robustness, and generalisation. A meta-learner, logistic regression further refines classification performance. This research proposes a stacking-based deep ensemble model for fake review detection, evaluated on an Amazon fake review dataset. It demonstrates improved accuracy compared to traditional machine learning and individual deep learning models.

The rest of the paper is structured as follows: Section 2 reviews related work, Section 3 details the proposed model, Section 4 presents experimental results, Section 5 discusses implications, and Section 6 concludes with future directions.

II. Related Work

Fake review detection has evolved significantly from traditional machine learning approaches to deep learning and ensemble-based models. Early research primarily focused on feature engineering, where linguistic, behavioural, and metadata-based attributes were manually extracted to classify deceptive opinions. Mewada et al. [7] examined traditional methods such as Support Vector Machines (SVM), Decision Trees, and Logistic Regression. Their study found that while these models successfully detected fake reviews, they suffered from high false-positive rates and limited generalisation due to the scarcity of large-scale labelled datasets. They also

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highlighted the challenges of relying on handcrafted features, which often fail to capture the nuanced patterns in deceptive reviews. Similarly, Mewada and Dewang [8] divided the techniques for fake review detection into three categories: spam review detection, spammer detection, and spammer-group detection. It was observed in this study that those methods, which used the feature of n-gram modelling, sentiment polarity analysis, and lexical richness focused on some specific domains, were effective. Nevertheless, these features did not provide strength in the context of broad datasets because their features were mostly compositional.

To address the gap that these older techniques provided, deep learning models are increasingly examined for automated feature extraction and further classification accuracy improvement. Mohawesh et al. [4] integrated LSTM and RoBERTa into a transformer architecture. Thus, it was suggested that capturing complex relationships between texts improves the detection of misleading reviews. The tests on known databases like OpSpam and Deception showed improvement in accuracy. Nonetheless, RoBERTa demonstrated those issues when capturing long-lived dependencies due to its fixed-length attention mechanism. This only demonstrates the lack of contextual relationships that can be provided using long sequences; thus, this necessitates a hybrid ensemble approach. Similarly, Mewada et al. [9] integrate sentiment-aware embeddings with BERT-based contextual representations. It effectively identifies fake reviews by analysing exaggerated sentiment polarity, but it lacks generalisation across different review platforms. Addressing another challenge, Gutierrez-Espinoza et al. [10] introduced a multi-modal deep learning framework incorporating text, images, and metadata. While their model improved fake review detection by leveraging cross-modal feature learning, it required large-scale annotated datasets, making it computationally expensive for real-world applications.

Apart from deep learning, ensemble learning techniques have shown significant promise in enhancing detection accuracy by combining multiple classifiers. Gutierrez-Espinoza et al. [11] demonstrated that ensemble methods like Bagging, Boosting, and Stacking consistently outperform single classifiers in handling noisy and imbalanced datasets. Hegde et al. [12] also introduced a hybrid ensemble learning model that integrates deep learning and machine learning classifiers. This approach combined word embedding-based deep neural networks with feature-based classifiers, significantly improving classification accuracy on Yelp datasets. However, this study lacked a meta-learning component, which could further optimise predictions. In another study, Mewada et al. [13] proposed a self-training hybrid model that combined S3SVM, SSDT, and USKMW to classify fake reviewers based on behavioural and content-based analysis. This method did not include ensemble stacking, which could enhance detection performance further and achieve even greater precision and recall scores. Mani et al. [14] explore Random Forest, Gradient Boosting, and Neural Networks ensemble models and proved that having a single ensemble feature base structure significantly increases classification performance, especially with imbalanced datasets, the primary problem in real-life situations.

Along with these improvements, the problem of fake review detection still possesses problems like data imbalance problems, cold start problems, and high computational costs. Most approaches are not generalisable across more than one dataset, or else they are extremely expensive if these problems are tried in the real world. With deep learning being the current trend on the rise, our stacking-based deep ensemble model hopes to solve these problems by combining multiple base classifiers with meta classifiers. As a result, combining a few lightweight models with a metaclassifier can guarantee high performance while at a low cost. This makes the solution for fake review detection robust, adaptive, and scalable. The gap that remains in the literature is meshed frameworks with high accuracy, high efficiency, and broader generalisation.

III. Methodology

A. Problem Definition

Deceptive opinion detection involves extracting fraudulent reviews and opinions from various sites. These reviews of services, products, or even people are intentionally made to elevate or depress their intended value falsely. The crux of the problem is to ascertain true and false reviews reliably. This task can be treated as a binary classification problem where a review can be classified as deceptive or non-deceptive. Non-deceptive classification is effective for formulating these problems due to its ease and the ease with which genuine and non-genuine reviews can be classified. Such a structure enables simple understanding and analysis of the outcomes, which renders it useful for actual problems. Formally, given a set of reviews $R = \{r1,r2,\ldots,rn\}$, each review ri will have an associated label yi, where yi $\{0,1\}$. Here, $yi = 0$ means the review is non-deceptive, while $yi = 1$ indicates it is a deceptive review. The aim is to estimate a mapping function $f: R \rightarrow Y, Y = \{0,1\}$ for the classifier to be able to guess the label yi with unseen reviews successfully. This is done by training a model to optimise classification performance measurement criteria like accuracy, precision, recall, F1 score, etc.

B. Dataset and Preprocessing

As mentioned earlier, this research uses Amazon reviews with a total of 21,000 reviews, out of which 10,500 are authentic reviews while 10,500 are fabricated. This dataset is accessible and commonly used in review fraud detection studies to increase reproducibility and comparability with other reviews. One caveat is that the data might be skewed by some biases, such as a particular type of product being reviewed more often than others or reviews having a large range of lengths, which might affect the model performance. Regardless of the constraints, the various attributes of the dataset, which contains multiple categories of products, increase the model's generalisation capability.

The following steps have been taken to preprocess the quality of the input data to enhance its quality. Initially, data housekeeping is carried out by eliminating HTML tags, special characters, and additional spaces to produce a clean document structure. This step aims to help reduce noise and maintain consistency throughout the text data. After this, the text undergoes tokenisation, where any sentences are split into single tokens, usually words or sub-words. This allows processes to run with greater efficiency. Broadening the window even further, stop words such as "the," "is," and "and", which hold little meaning to the overall statement, tend to get removed. These steps help provide relief in artificial review detection models.

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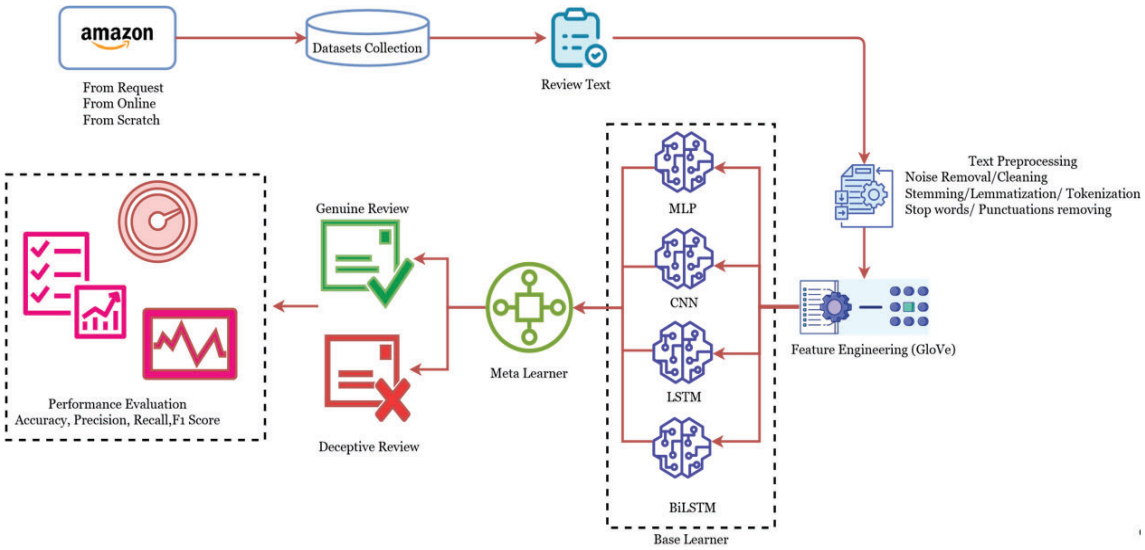


Fig. 1. Proposed model pipelining

Global Vectors for Word Representation (GloVe) uses embeddings to transform written text into numbers. GloVe was selected over other word embeddings such as Word2Vec and FastText because GloVe derives global statistical value from the whole corpus, meaning it pays attention to bigger semantic and syntactic relationships between words. Each word is allocated to a specific position in a high-dimensional vector space, which allows the model to handle these relationships better in fake review detection.

C. *Proposed Stacking-Based Deep Ensemble Model*

Stacking ensemble learning enhances classification accuracy by integrating multiple base models in a hierarchical approach. Unlike traditional methods such as bagging or boosting, which rely on averaging or weighted voting, stacking trains diverse base models independently and uses their predictions as inputs for a meta-learner. This meta-learner determines the optimal combination of predictions, improving classification performance, model generalisation, and robustness. The proposed model, as shown in Figure 1, consists of two levels. The first level includes multiple base models (MLP, CNN, LSTM, and BiLSTM), each capturing distinct textual features. MLP transforms high-dimensional feature maps into compact vector representations, CNN extracts hierarchical text patterns, LSTM retains long-term dependencies in sequential data, and BiLSTM improves contextual understanding by processing text in forward and backward directions. These models generate predictions, which are then passed to the second level.

The second level consists of a meta-learner that refines the final classification decision. Instead of simple averaging, the meta-learner assigns optimal weights to each base model's prediction to enhance accuracy. Logistic regression was chosen as the meta-learner due to its simplicity, interpretability, and efficiency in combining predictions from diverse models. Less sophisticated meta-learners, such as logistic regression, do not suffer from overfitting, as they balance the influence of each model rather well. Alternatively, error correction and robust classifications can be accomplished through fully connected networks serving as meta-learners, which capture complex relationships amongst base model outputs.

The entire procedure guarantees classification at a very high level. Each base model uses distinct labelled data sets for training. Grid search and Bayesian optimisation techniques are employed to improve performance and prevent overfitting. These approaches optimise key parameters such as learning, dropout, and batch size rates for optimal base model performance. Base models give predictions on a validation dataset they have not seen, and then these predictions are used to create features for the meta-learner. The meta-learner is trained to combine the predictions to improve the overall classification optimally. The stacking-based deep ensemble model efficiently merges multiple architectures and is useful in solving intricate classification problems like fake review detection.

IV. Experimental Results And Analysis

A. *Performance Comparison with Baseline Models*

To assess the performance of the suggested stacked model, we also evaluated the results for baseline models such as MLP, CNN, LSTM, and BiLSTM. Accuracy (Acc), Precision (Pre), Recall (Rec) and F1-score [15][16] are the evaluation metrics that measure the performance of the proposed and existing baseline models. The comparative performance results of different models are included in Table 1. The results revealed that the stacked model surpasses the performance of baseline models in all measured criteria.

TABLE I. Performance Metrics Of Different Models (*Proposed Model)

Model	Acc (%)	Pre (%)	Rec (%)	F1 (%)
MLP	80.00	75.00	85.00	80.00
CNN	83.11	80.03	81.82	80.90
LSTM	86.10	81.25	88.64	84.78
BiLSTM	87.13	86.05	84.12	85.06
Stacked Model*	90.04	85.96	91.31	88.89

The ensemble method is more powerful here because the four models work more powerfully. For example, feature representations are captured in high-dimensional space by MLP, while CNN captures text patterns at different levels of the hierarchy. LSTM and BiLSTM models capture sequential and contextual information. Taking these suggestions together helps the stacked model overcome the shortcomings of

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individual models and, by doing so, achieve greater robustness and generalisation. Furthermore, the meta-learner divides the importance of each base model to maximise classification, accuracy and reliability. As shown in Figure 2, these opportunities prove that stacking-based ensemble models must combine different architectures to enhance the system's performance.

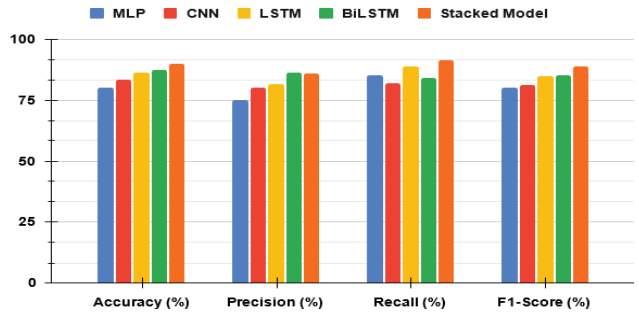


Fig. 2. Performance Metrics of the Stacked model with other deep learning models

B. *Ablation Study: Impact of Individual Base Models*

To determine the importance of the baseline models, we performed an ablation study by deleting individual components of the stacked model and checking for performance loss. You can find the results in Table II. The results show that the highest performance degradation was recorded when the BiLSTM component was removed, which indicates that its contribution towards feature extraction is extremely high. This is because BiLSTM can capture sequential dependencies in both forward and backward directions, which allows for a better understanding of the contextual relationships within the text. By processing sequences bidirectionally, BiLSTM can identify patterns and nuances that would be missed by unidirectional models such as LSTM, which is why it is an important component of the stacked model. Removing other base models, such as MLP, CNN, or LSTM, causes a smoother degradation of performance, which attests to the complementary role of all parts in an ensemble framework.

TABLE II. ABLATION STUDY: IMPACT OF REMOVING BASE MODELS (* PROPOSED MODEL)

Model	Acc (%)	Pre (%)	Rec (%)	F1 (%)
Without MLP	89.00	87.00	90.50	88.72
Without CNN	88.20	86.50	89.80	88.12
Without LSTM	87.50	85.80	89.00	87.36
Without BiLSTM	86.80	84.90	88.30	86.56
Stacked Model* (All)	90.04	88.50	91.31	89.89

V. CONCLUSION AND FUTURE WORK

This research analysed the effectiveness of deeply interdisciplinary models like MLP, CNN, LSTM, BiLSTM and the stacked ensemble model towards sequence classification. Of these methods, the ensemble model performs the best with an accuracy of 90.04, which is higher than all the individual models. The ablation study presented above demonstrated the importance of BiLSTM, given its ability to exploit bidirectional contextual dependencies.

To improve the results, future work could employ more sophisticated transformer-based BERT or GPT architectures for better comprehension. Multi-modal approaches incorporating text, images, or audio could enrich learning. Automated hyperparameter tuning techniques, such as Bayesian optimisation, and model compression methods, like quantisation, could improve efficiency and scalability. Expanding the model's application to domains like time-series forecasting or anomaly detection would also broaden its utility. Improving model explainability and fairness is crucial for real-world deployment, ensuring reduced bias and increased interpretability. In distributed environments, federated learning could enhance privacy and security, while lightweight model versions could enable real-time deployment. By addressing these areas, future research can advance deep learning models' accuracy, reliability, and practical usability in complex classification tasks.

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